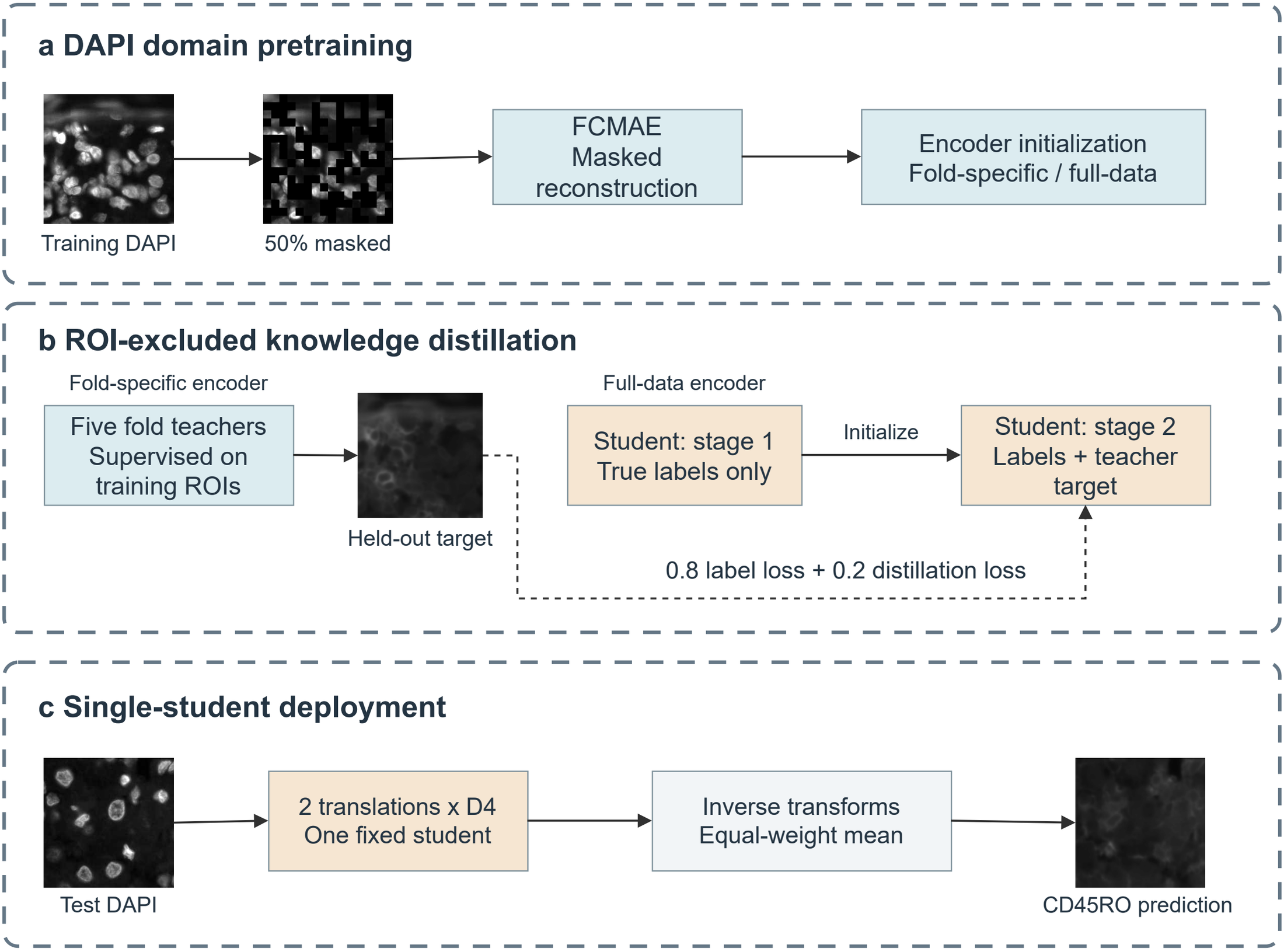

a DAPI domain pretraining
Training DAPI
50% masked
FCMAE
Masked
reconstruction
Encoder initialization
Fold-specific / full-data
b ROI-excluded knowledge distillation
Fold-specific encoder
Five fold teachers
Supervised on
training ROIs
Held-out target
Full-data encoder
Student: stage 1
True labels only
Initialize
Student: stage 2
Labels + teacher
target
0.8 label loss + 0.2 distillation loss
c Single-student deployment
Test DAPI
2 translations x D4
One fixed student
Inverse transforms
Equal-weight mean
CD45RO prediction